Synchronization concept for a deterministic Bunch-to-Bucket transfer for FAIR

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Abstract

The FAIR accelerator facility is expected to be one of the largest ion research centres worldwide. In the frame of this project, the synchrotron SIS18 will serve as an injector for the SIS100, currently under construction. A deterministic Bunch-to-Bucket synchronization procedure is one of the key features of the beam transfer system between SIS18 and SIS100. The Bunch-to-Bucket transfer for FAIR must be scalable to any type of ion which is accelerated in the SIS18.

Two extreme cycles are taken into account in the frame of this research: protons up to an energy of 4 GeV and uranium ions U^{28+} up to an energy of 196 MeV/u at transfer flattop in the SIS18. A dedicated low level radio frequency (LLRF) solution, based on local signal generation and high speed optical data transfer is being developed at GSI. This LLRF topology is expected to fulfil all the required functions of the Bunch-to-Bucket synchronization and transfer, from the phase advance measurement to the possible adiabatic phase steering and further control procedures.

Introduction : accelerated particles, rf bucket and synchronization

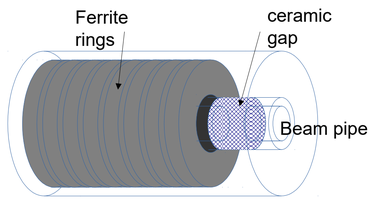


Figure 1: a ferrite cavity

Particles in a synchrotron are gathered into bunches within rf potential wells of the electromagnetic field, generated in the rf cavities (see fig. 1). The phase space representation of these potential wells is called rf bucket (see fig. 2).

In the frame of the FAIR project, it is intended to perform fast extractions of bunched beams from a source synchrotron to a target synchrotron. Such a transfer scheme is called Bunch-to-Bucket transfer.

In order to enable the capture of the incoming bunches by the target synchrotron rf system, a synchronization step prior to the transfer is required (see fig. 3) [1].

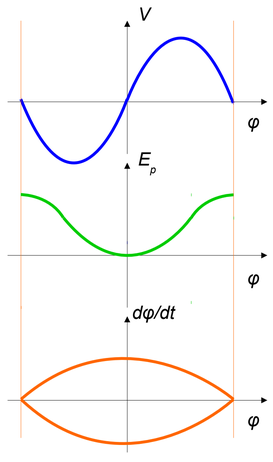


Figure 2: a sinus rf voltage, with the corresponding potential well and rf bucket (stationary case)

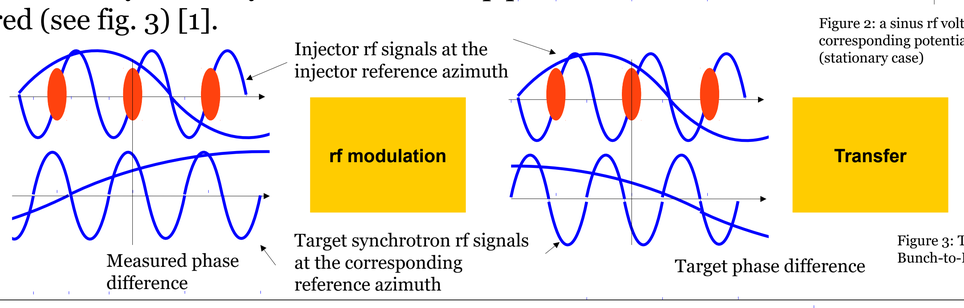


Figure 3: The principle of the Bunch-to-Bucket transfer

The FAIR acceleration chain : a cable length issue

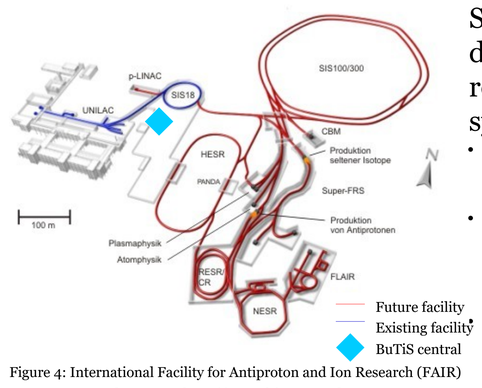


Figure 4: International Facility for Antiproton and Ion Research (FAIR) at GSI, Darmstadt. Source: http://www.fair-center.de

Synchronizing the rf generation systems of two different synchrotrons within one machine cycle requires fast data exchange between all the synchronization actors. This takes into concern:

- Connections within a supply area (< 100 m, rf signals and short range optical fibres).
- Connections between rf supply rooms and/or kicker control rooms (< 500 m, mono-mode optical fibres, see fig. 5).
- Central connections to the BuTiS station and to the White Rabbit network (optical Ethernet, see fig. 4) [2].

The modular low level hardware implementation concept deployed at GSI enables local synchronous reference signal generation all over the accelerator facility.

The concept for a deterministic Bunch-to-Bucket transfer for FAIR makes use of it and relies on three main functions.

- Phase Advance Prediction (PAP): this function aims at time-stamping the phase difference and advance measurement, consistently with the BuTiS reference.
- Signal Reproduction (SR+PCM): this function reproduces the rf signals of the target synchrotron wherever it is implemented (see next section of this poster).
- Phase Shift (PSM+OSA): this function fulfils the active phase or frequency modulation of the reference rf signals in open loop (see last section of this poster).

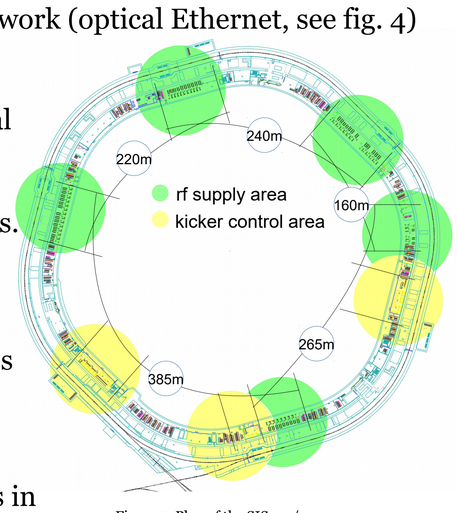


Figure 5: Plan of the SIS100/300

Local reference signal generation and re-synchronization

The functions mentioned above are going to be implemented on the basis of a new Multi-purpose Hardware Unit, developed in cooperation with the IES institute (Technische Universität Darmstadt).

Fig. 6 shows the standard Bunch-to-Bucket topology for one supply room as defined in [3].

In order to validate the concept, developed for FAIR, experiments are going to be performed on the two main mechanisms, involved in the synchronization procedure. The first series of experiments will aim at performing phase advance time-stamping and the local target synchrotron rf signal reproduction. Combining these two functions enable to performed a fully operational passive synchronization, called frequency beating. Fig. 7 shows the hardware topology related to this experiment.

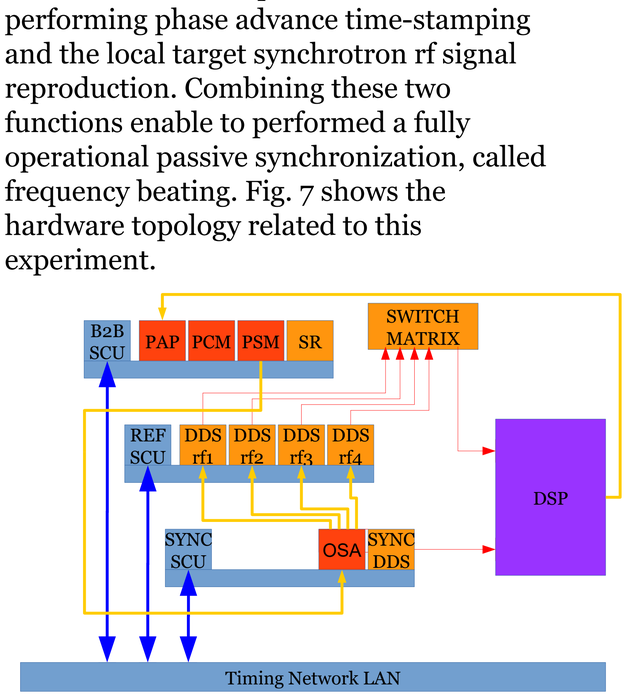


Figure 6: Functional block representation of the standard LLRF hardware implementation for a deterministic Bunch-to-Bucket transfer for FAIR.

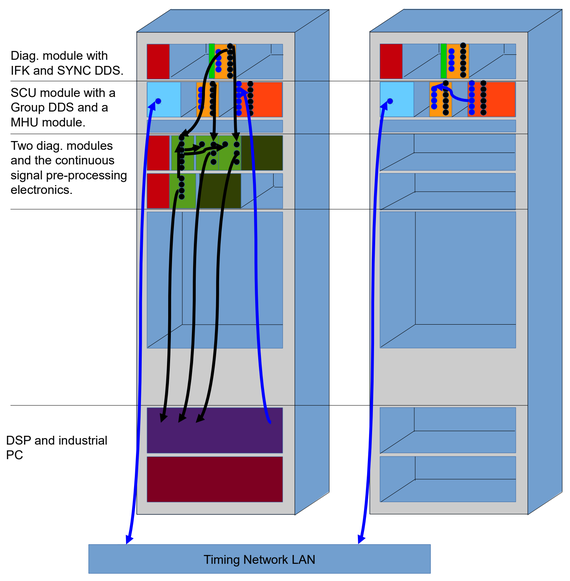


Figure 7: Hardware implementation of the re-synchronization function on the rack planning level.

Table 1: the hardware modules of Fig. 6.

Module	Function
DDS	reference rf signal generation
DSP	phase difference measurement
SWITCH MATRIX	rf signal distribution
SCU	interface to Timing Network
SR	rf signal reproduction (on DDS basis)
PSM	reference rf signal phase or frequency modulation
PAP	phase difference and advance measurement time stamping (link to BuTiS)
OSA	Synchronous DDS correction and inter-room real time data sharing
PCM	SR re-synchronization

Adiabatic phase steering simulation

The second proof, which is required in the frame of this project, consist in applying a phase shift, consisting of an active adiabatic phase modulation of the reference rf signal.

The dynamic of particles in a synchrotron can be characterized with respect to the centre of the rf bucket in which it is trapped according to Eq. (1).

$$\ddot{\varphi} + \omega_s^2\left(\sin\varphi - \sin\varphi_s\right) = 0 \qquad (1)$$

By modulating the cavity driving signals in a controlled way, one can shift the rf buckets within the source synchrotron to compensate the delays related to the beam transfer. This operation must remain adiabatic [4], i.e.:

$$\epsilon = \frac{1}{\omega_s^2}\left|\frac{\mathrm{d}\,\omega_s}{\mathrm{d}\,t}\right| \ll 1 \qquad (2)$$

Fig. 8 and 9 represent results of simulation for such an open loop phase shift by 180 degrees within 100 ms for protons at extraction energy (26 GeV) for h=84 at PS.

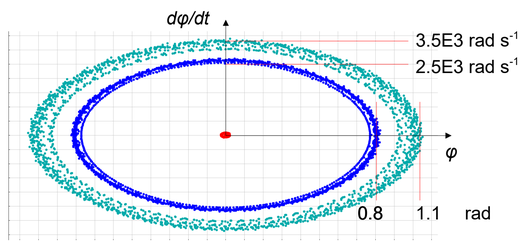


Figure 9: Particle trajectory in phase space. The blue trace shows the particle dynamic for an emittance of 0.35 eVs. The light blue trace for 0.4 eVs.

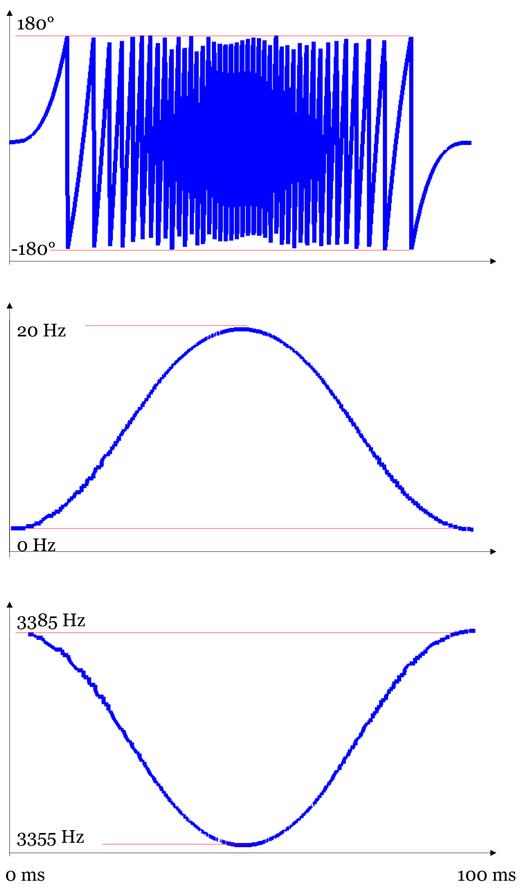


Figure 8: Phase modulation, frequency modulation and related synchrotron frequency for a simulated phase steering (at PS).

References:
[1] T. Ferrand, Synchronization of Synchrotrons for bunch-to-bucket Transfer, 2015.
[2] J. Serrano & al., The White Rabbit Project.
[3] T. Ferrand and J. Bai, Concept of the FAIR Bunch-to-Bucket transfer system, Internal Documentation 2016.
[4] A. W. Chao and M. Tigner, Accelerator Physics and Engineering, 3rd printing, 2009.

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